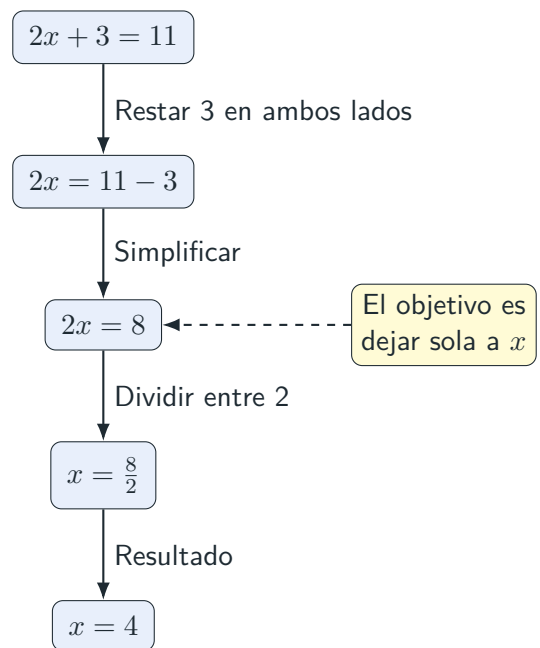


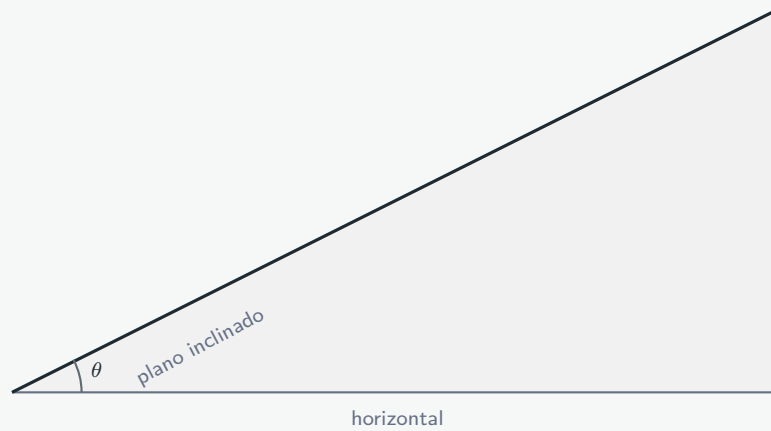
Diagramas TikZ

Ejemplo con plano inclinado y overlays

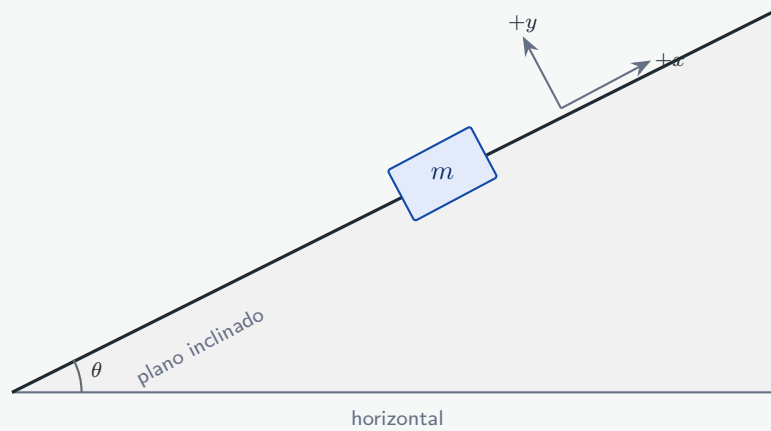




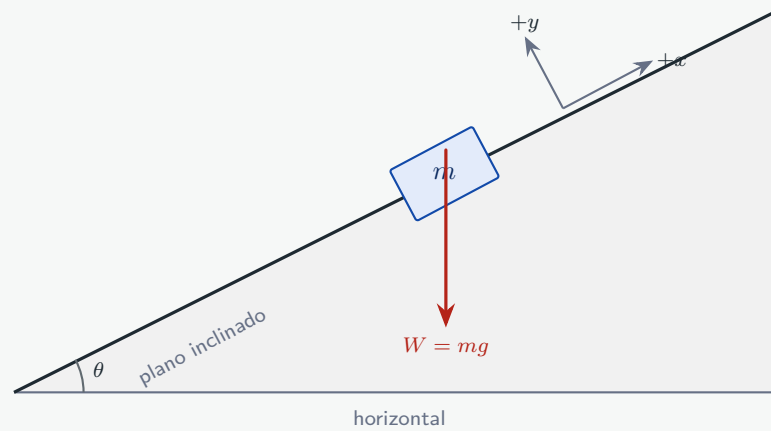
Sistema físico



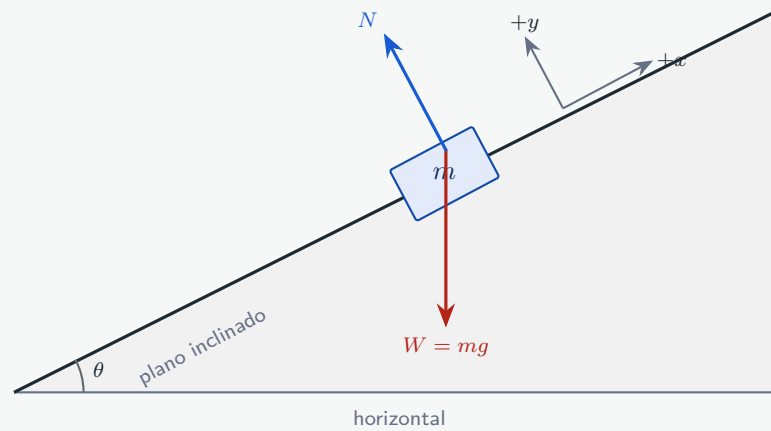
Sistema físico



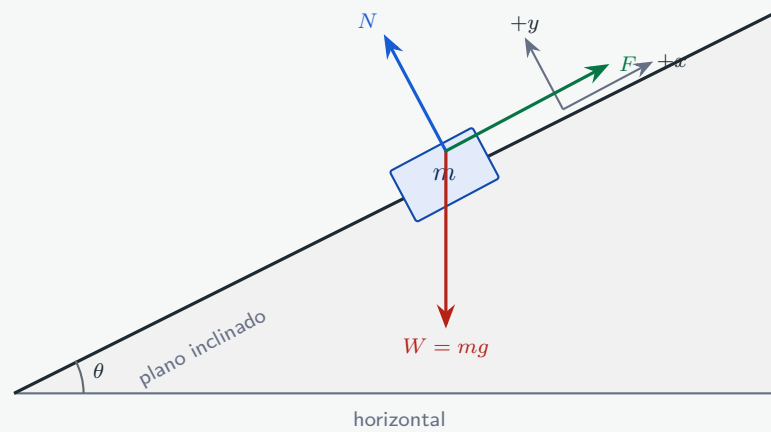
Sistema físico



Sistema físico



Sistema físico

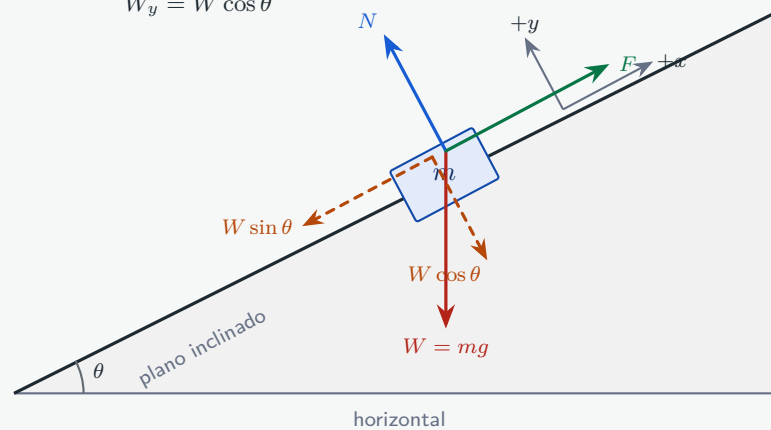


Sistema físico

Descomposición del peso:

$$W_x = W \sin \theta$$

$$W_y = W \cos \theta$$

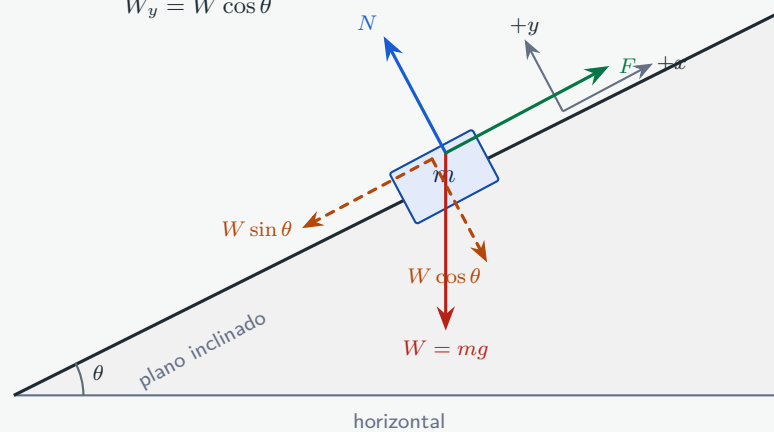


Sistema físico

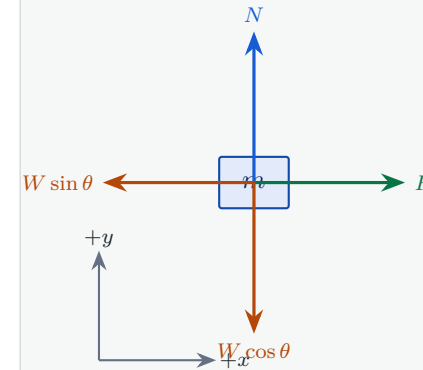
Descomposición del peso:

$$W_x = W \sin \theta$$

$$W_y = W \cos \theta$$



DCL: cuerpo libre

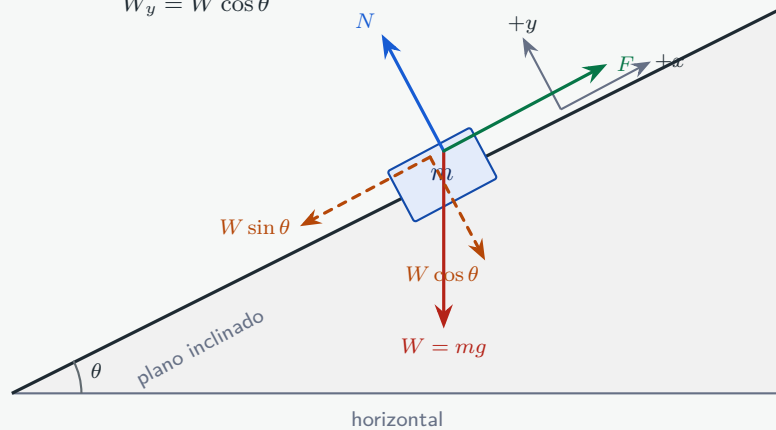


Sistema físico

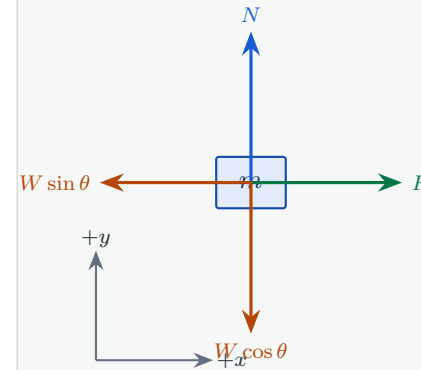
Descomposición del peso:

$$W_x = W \sin \theta$$

$$W_y = W \cos \theta$$



DCL: cuerpo libre



Modelo dinámico

$$\sum F_x = F - W \sin \theta$$

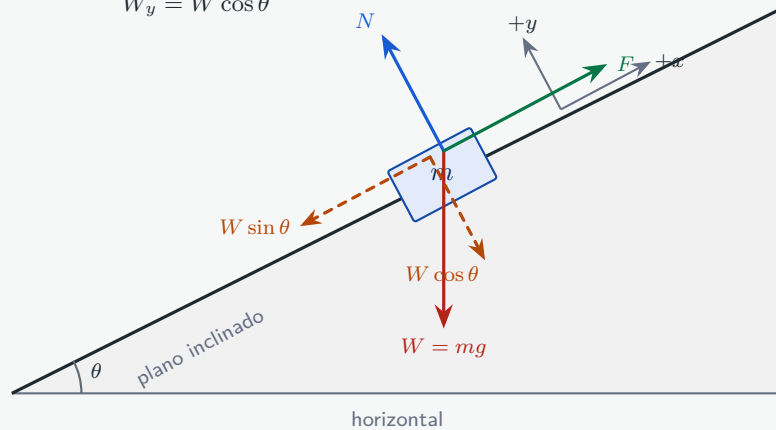
$$\sum F_y = N - W \cos \theta = 0$$

Sistema físico

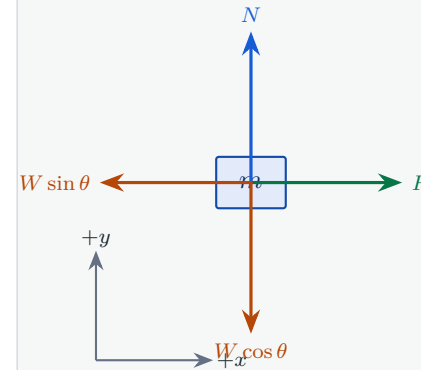
Descomposición del peso:

$$W_x = W \sin \theta$$

$$W_y = W \cos \theta$$



DCL: cuerpo libre



Modelo dinámico

$$\sum F_x = F - W \sin \theta$$

$$\sum F_y = N - W \cos \theta = 0$$

Como $W = mg$:

$$ma = F - mg \sin \theta$$

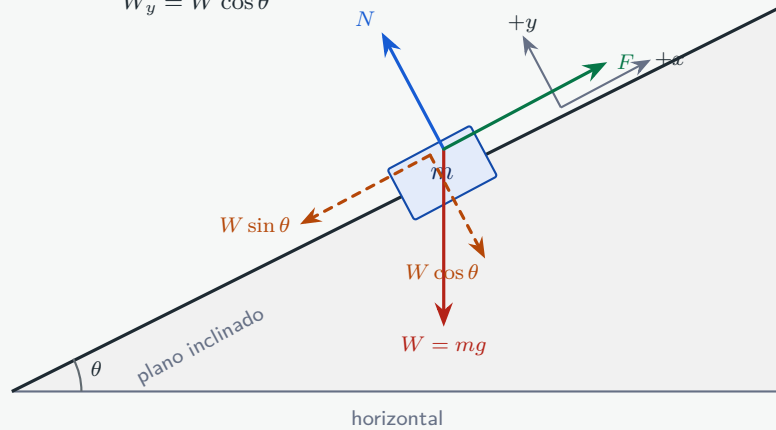
$$a = \frac{F - mg \sin \theta}{m}$$

Sistema físico

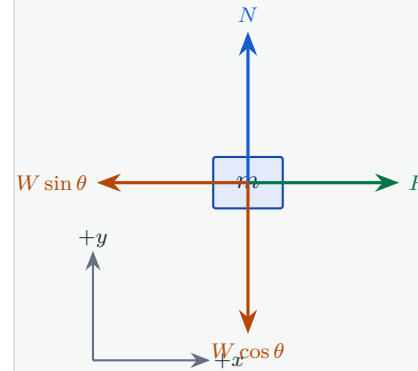
Descomposición del peso:

$$W_x = W \sin \theta$$

$$W_y = W \cos \theta$$



DCL: cuerpo libre



Modelo dinámico

$$\sum F_x = F - W \sin \theta$$

$$\sum F_y = N - W \cos \theta = 0$$

Como $W = mg$:

$$ma = F - mg \sin \theta$$

$$a = \frac{F - mg \sin \theta}{m}$$

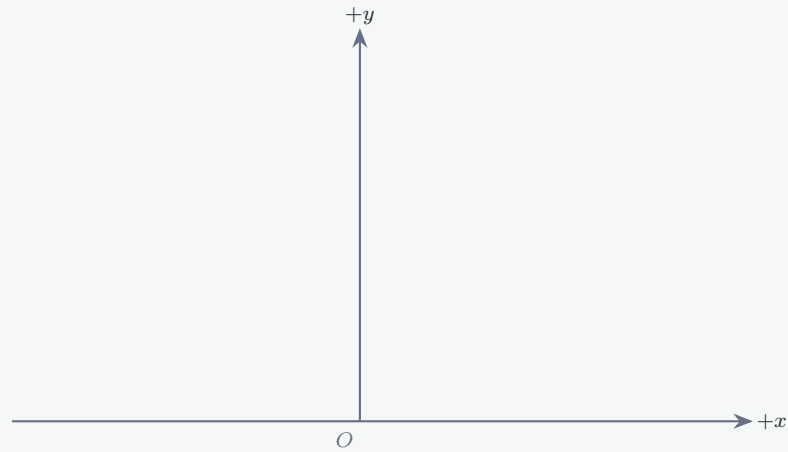
Cinemática

Si $v_0 = 0$ y $t = 3$ s:

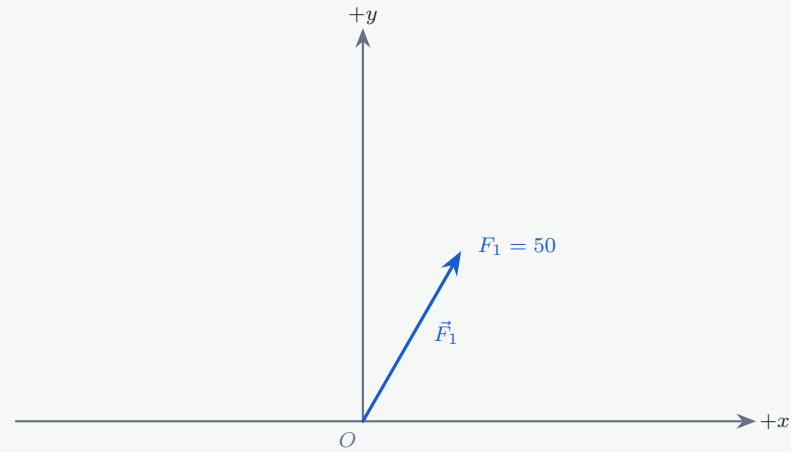
$$x = \frac{1}{2}a(3)^2$$

$$v_f = a(3)$$

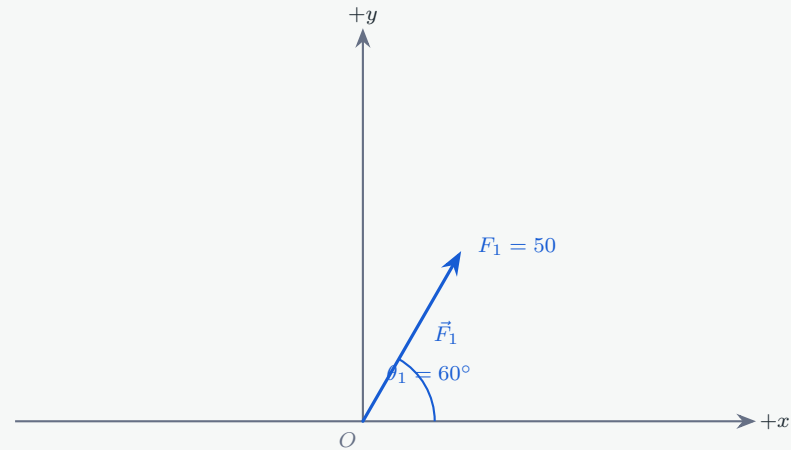
Sistema vectorial



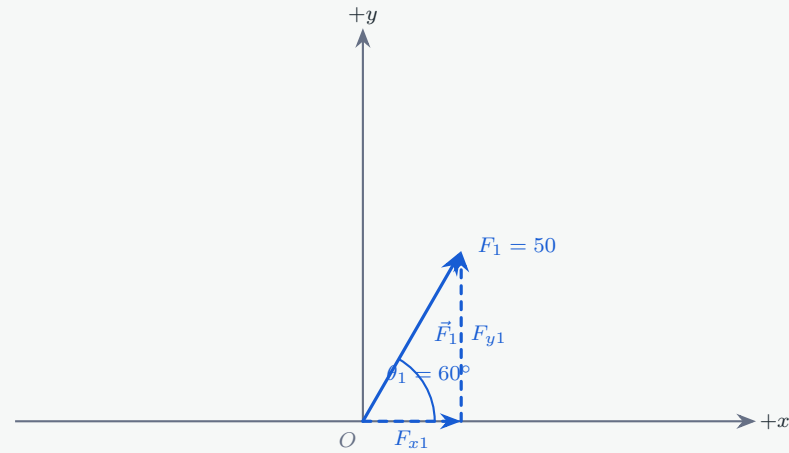
Sistema vectorial



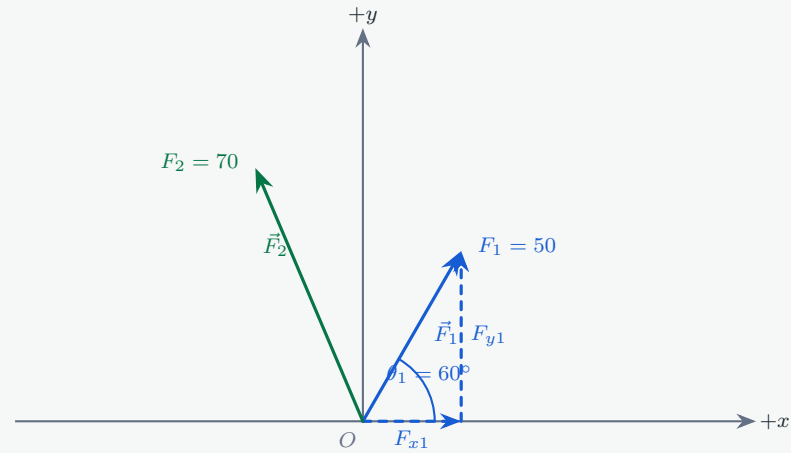
Sistema vectorial



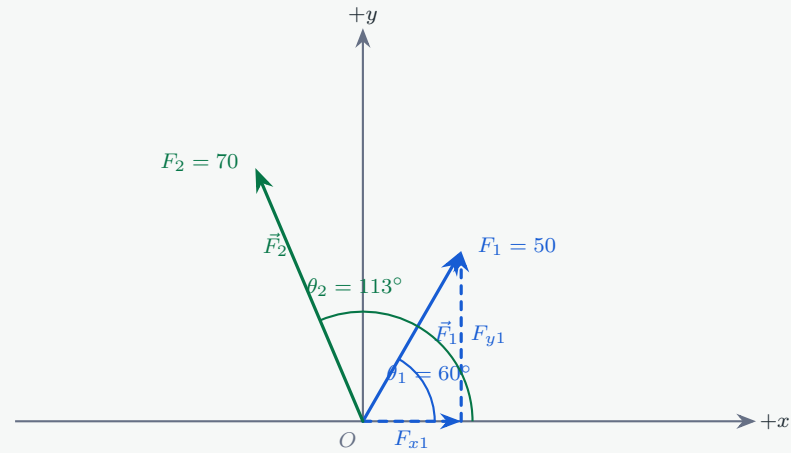
Sistema vectorial



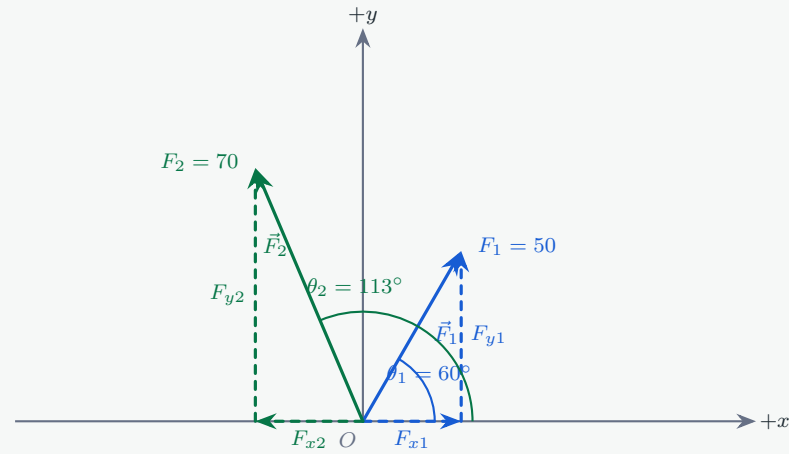
Sistema vectorial



Sistema vectorial



Sistema vectorial



Sistema vectorial

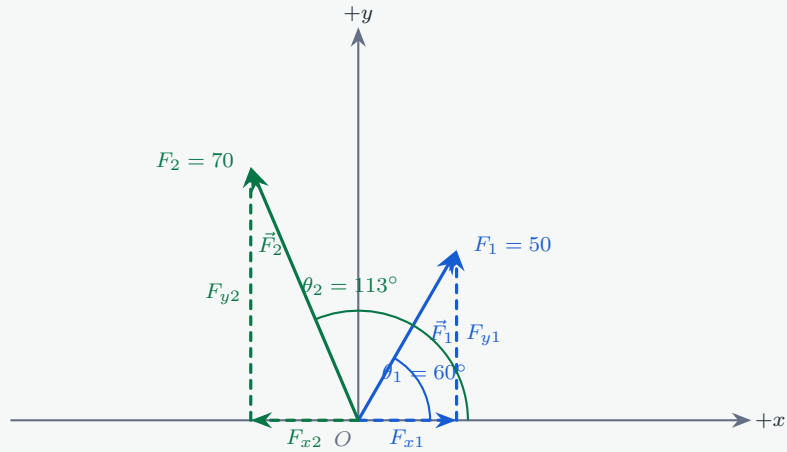


Tabla de componentes

Vector	F_x	F_y
$\vec{F}_1$	$F_1 \cos \theta_1$	$F_1 \sin \theta_1$
$\vec{F}_2$	$F_2 \cos \theta_2$	$F_2 \sin \theta_2$

Sistema vectorial

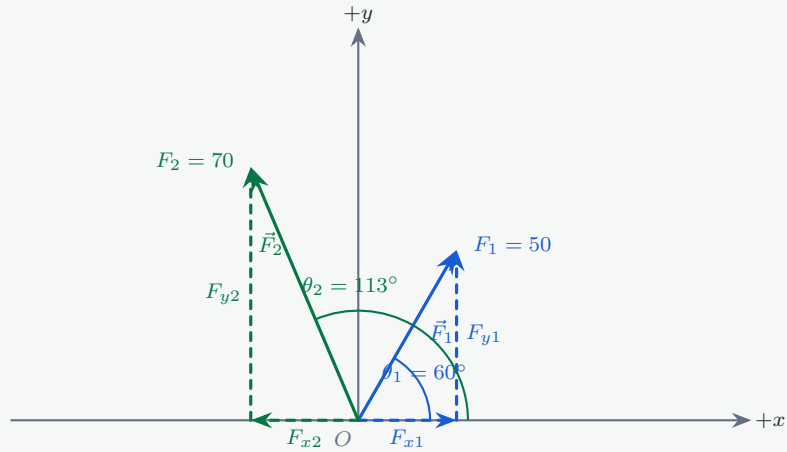


Tabla de componentes

Vector	F_x	F_y
$\vec{F}_1$	$F_1 \cos \theta_1$	$F_1 \sin \theta_1$
$\vec{F}_2$	$F_2 \cos \theta_2$	$F_2 \sin \theta_2$

Para $\vec{F}_1$:

$$F_{x1} = 50 \cos(60^\circ) = 25.00$$

$$F_{y1} = 50 \sin(60^\circ) = 43.30$$

Sistema vectorial

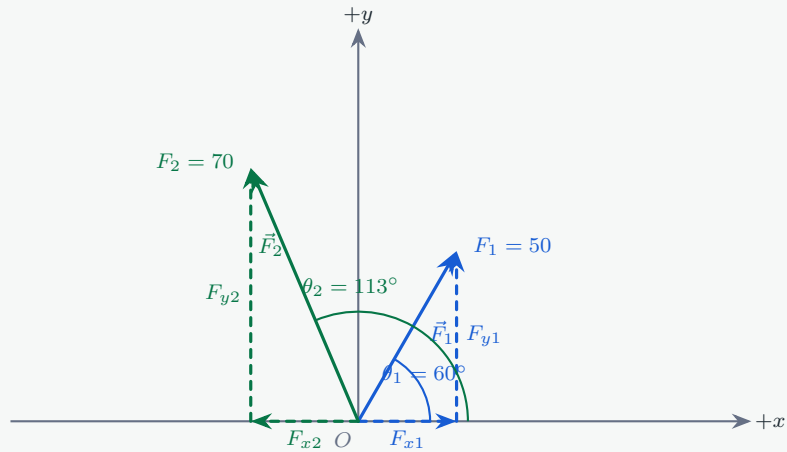


Tabla de componentes

Vector	F_x	F_y
$\vec{F}_1$	$F_1 \cos \theta_1$	$F_1 \sin \theta_1$
$\vec{F}_2$	$F_2 \cos \theta_2$	$F_2 \sin \theta_2$

Para $\vec{F}_1$:

$$F_{x1} = 50 \cos(60^\circ) = 25.00$$

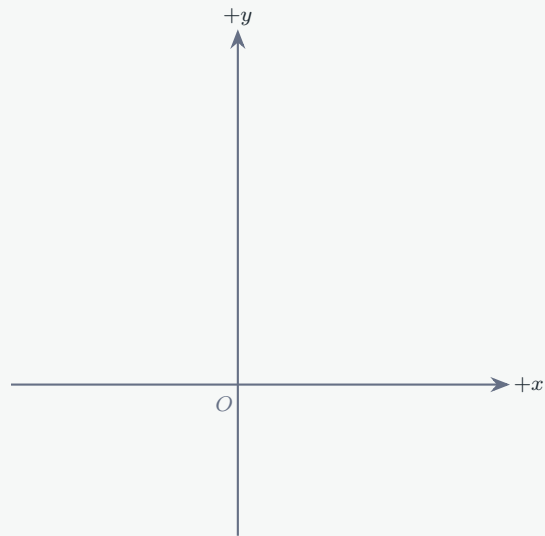
$$F_{y1} = 50 \sin(60^\circ) = 43.30$$

Para $\vec{F}_2$:

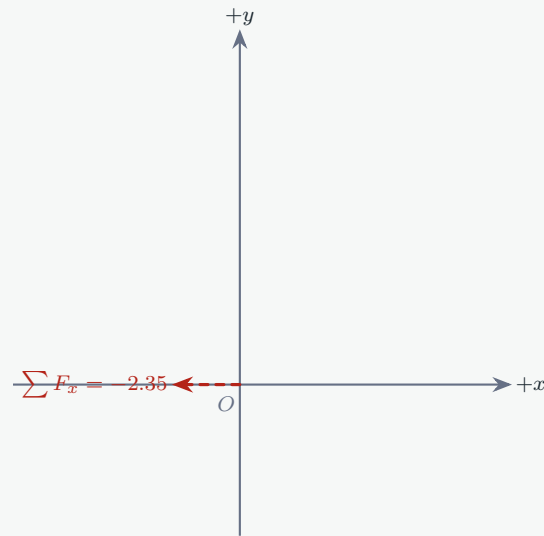
$$F_{x2} = 70 \cos(113^\circ) = -27.35$$

$$F_{y2} = 70 \sin(113^\circ) = 64.44$$

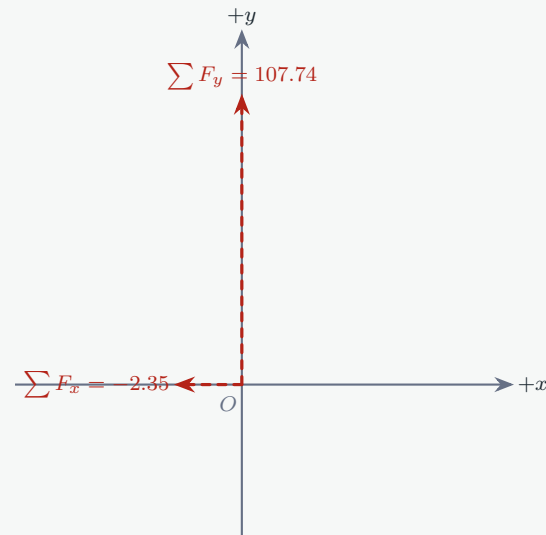
DCL de la resultante



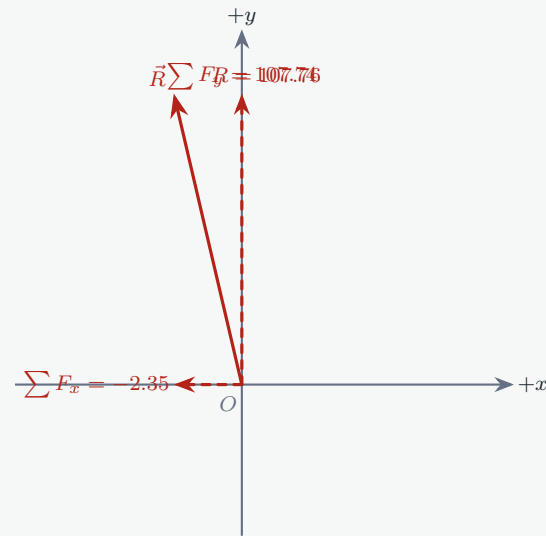
DCL de la resultante



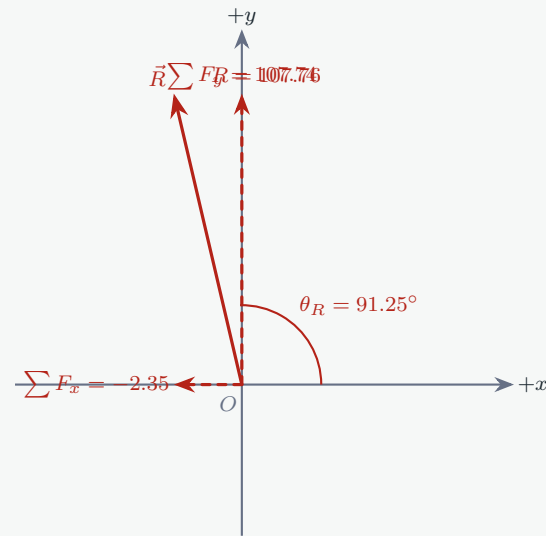
DCL de la resultante



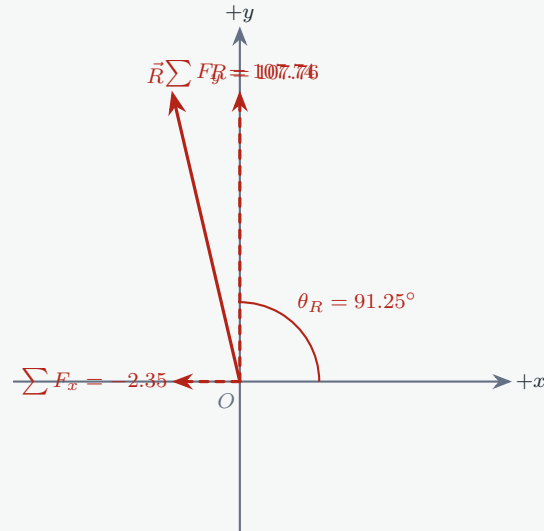
DCL de la resultante



DCL de la resultante



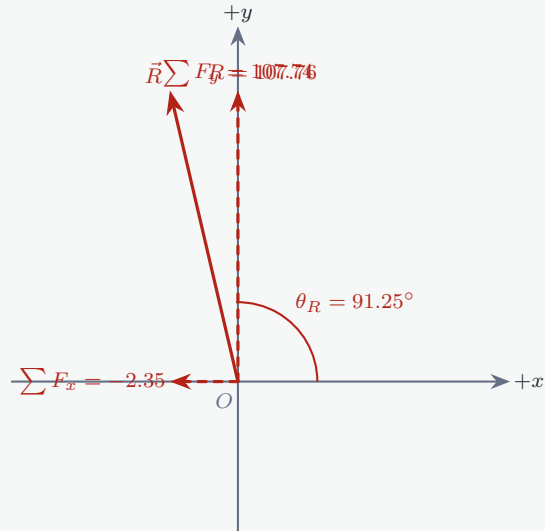
DCL de la resultante



La resultante queda casi vertical, ligeramente hacia la izquierda,
porque:

$$\sum F_x < 0, \quad \sum F_y > 0$$

DCL de la resultante



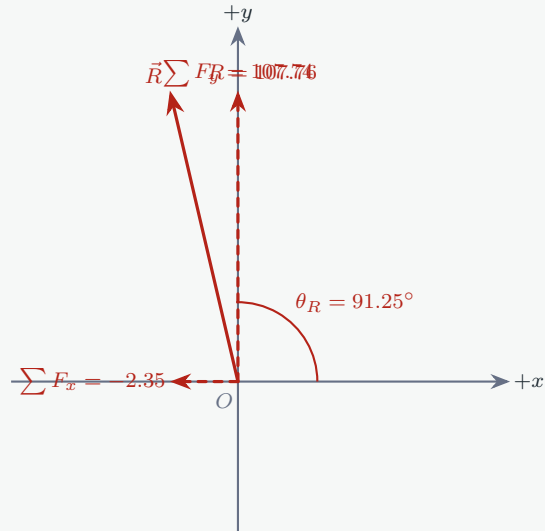
La resultante queda casi vertical, ligeramente hacia la izquierda, porque:

$$\sum F_x < 0, \quad \sum F_y > 0$$

Sumatorias

$$\begin{aligned} \sum F_x &= F_{x1} + F_{x2} \\ &= 25.00 + (-27.35) \\ &= \boxed{-2.35} \end{aligned}$$

DCL de la resultante



La resultante queda casi vertical, ligeramente hacia la izquierda, porque:

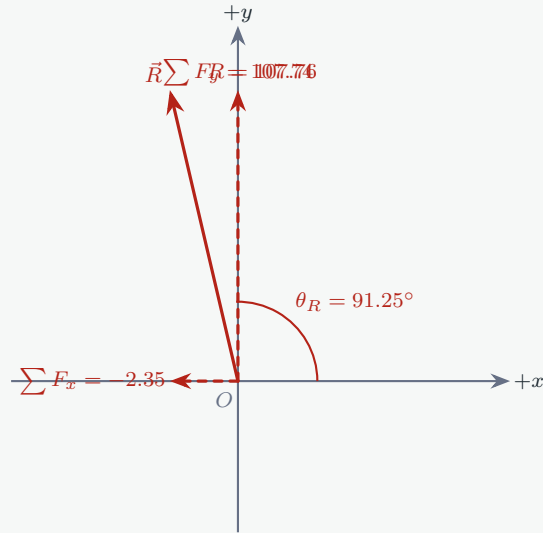
$$\sum F_x < 0, \quad \sum F_y > 0$$

Sumatorias

$$\begin{aligned}\sum F_x &= F_{x1} + F_{x2} \\ &= 25.00 + (-27.35) \\ &= \boxed{-2.35}\end{aligned}$$

$$\begin{aligned}\sum F_y &= F_{y1} + F_{y2} \\ &= 43.30 + 64.44 \\ &= \boxed{107.74}\end{aligned}$$

DCL de la resultante



La resultante queda casi vertical, ligeramente hacia la izquierda, porque:

$$\sum F_x < 0, \quad \sum F_y > 0$$

Sumatorias

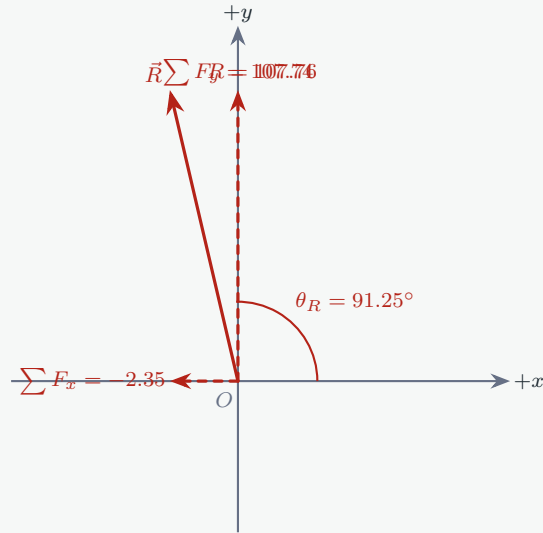
$$\begin{aligned}\sum F_x &= F_{x1} + F_{x2} \\ &= 25.00 + (-27.35) \\ &= \boxed{-2.35}\end{aligned}$$

$$\begin{aligned}\sum F_y &= F_{y1} + F_{y2} \\ &= 43.30 + 64.44 \\ &= \boxed{107.74}\end{aligned}$$

Vector resultante

$$\begin{aligned}R &= \sqrt{(\sum F_x)^2 + (\sum F_y)^2} \\ &= \sqrt{(-2.35)^2 + (107.74)^2} \\ &= \boxed{107.76}\end{aligned}$$

DCL de la resultante



La resultante queda casi vertical, ligeramente hacia la izquierda, porque:

$$\sum F_x < 0, \quad \sum F_y > 0$$

Sumatorias

$$\begin{aligned} \sum F_x &= F_{x1} + F_{x2} \\ &= 25.00 + (-27.35) \\ &= \boxed{-2.35} \end{aligned}$$

$$\begin{aligned} \sum F_y &= F_{y1} + F_{y2} \\ &= 43.30 + 64.44 \\ &= \boxed{107.74} \end{aligned}$$

Vector resultante

$$\begin{aligned} R &= \sqrt{(\sum F_x)^2 + (\sum F_y)^2} \\ &= \sqrt{(-2.35)^2 + (107.74)^2} \\ &= \boxed{107.76} \end{aligned}$$

$$\begin{aligned} \theta_R &= \text{atan2}(\sum F_y, \sum F_x) \\ &= \text{atan2}(107.74, -2.35) \\ &= \boxed{91.25^\circ} \end{aligned}$$